

Connect

Leveraging Large Language Models for Review Classification and Rating Estimation of Mental Health Applications

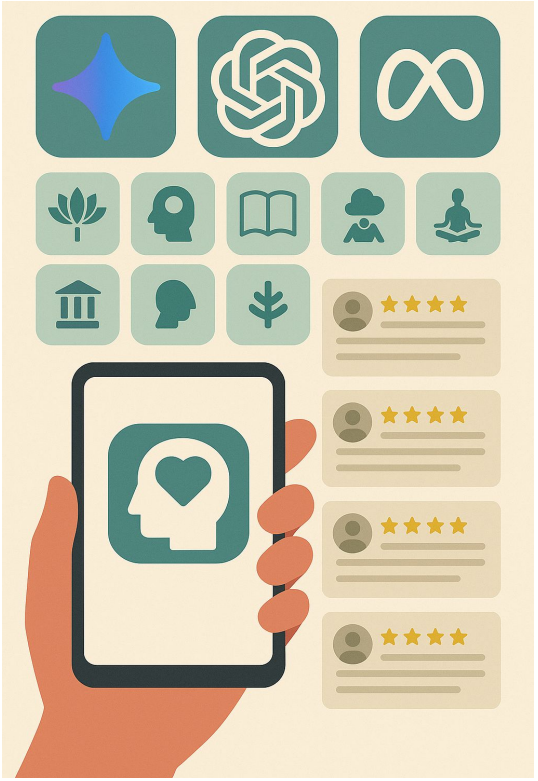
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Reasoning with Ratings

- Mental health **Apps** provide scalable support to individuals who lack access to traditional care.
- With over **10k** App available on the market, individuals rely on **user reviews to assess their quality and make decision**
- With the vision of building app recommendation system, we first explore if we can leverage Large Language Models (LLMs) to **accurately classify** user reviews in ratings into 1-5 stars?



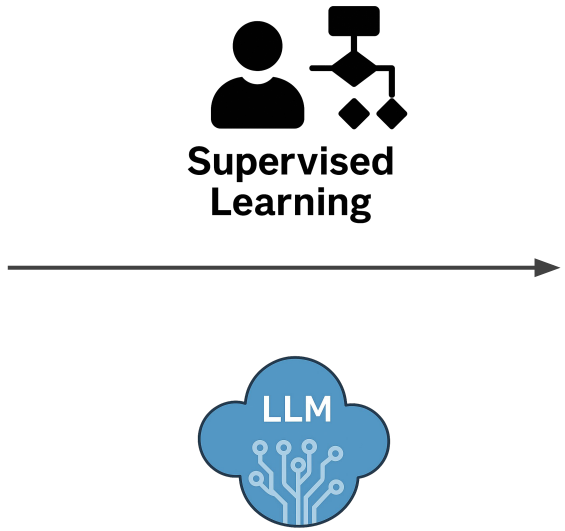
The Prediction Goal

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Benchmarking Large Language Models (LLMs) for Review Classification



Sample Review for "*7 Cups*" App



Predictions

Behind the Scenes

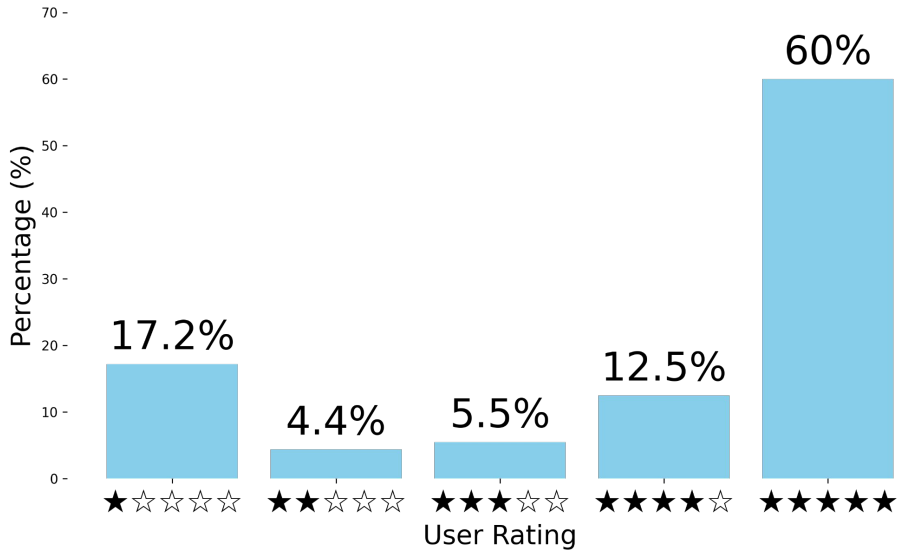
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Data Collection | Model Selection | Evaluation | Human Feedback

- **200,973 reviews**
- **73 popular mental health apps** (100k+ downloads)

Google Play Store

- 8 features (date, user reviews, ratings, business response, likes, etc.)
- PII Anonymized during text cleaning



Apps Rating Distribution: (J-curve)

Behind the Scenes

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Data Collection | **Model Selection** | Evaluation | Human Feedback

18 Supervised Learning

- Traditional ML Algorithms, Deep Learning

7 Self-Supervised LLM

- OpenAI, Google, Meta

Prompt: "You are an Expert APP RATER specializing in MENTAL HEALTH applications. Your task is to ACCURATELY CLASSIFY the review into one of the rating classes where 1 being the lowest and 5 being the highest. Only return a single number from Classes: [1, 2, 3, 4, 5] {individual user review}"

Zero-shot Learning

Behind the Scenes



Data Collection | Model Selection | **Evaluation** | Human Feedback

Metrics

- Precision
- Recall
- F1-score

Computed at the **individual** review level, then **aggregated** at the app level.

Behind the Scenes



Data Collection | Model Selection | Evaluation | Human Feedback

- Recruited **crowdsource** workers to examine **discrepancies**
  - cases where GPT-predicted ratings misclassified from original user ratings
- **Thematic analysis** used to investigate underlying patterns and reasoning

Grab and Go

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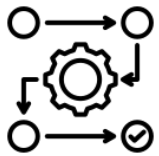
Feasible & Efficient:

- 80% accuracy from best supervised learning (BERT)
- 75% accuracy from best LLMs (GPT-4 and LLaMA 3.3 70B)



Subjectivity:

- **60%** of AMT workers preferred GPT's ratings over the original user ratings
- Suggest LLMs may have **less subjective measure** of user sentiment



Can LLMs reflect app store rankings?

- GPT-4 identified 8 out of the top 10 apps, showing strong ranking recall

Thank You



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